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Biology
Class 9

Sindh Textbook Board

(English Medium)

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Introduction to Biology

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Solving a Biological Problem

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Introduction to Biology

UNIT 1

1.1 Divisions and Branches of Biology

LESSON 1

1. Introduction

Biology is divided into many branches, each branch studies one part of living things. Biology is also linked with many other sciences, so it is called a multidisciplinary subject.

2. Important Branches of Biology

Branch	What it studies
Anatomy	The internal structure of the bodies of organisms.
Histology	The structure of tissues.
Immunology	The body's immune system and protection against diseases.
Pharmacology	The science of medicines; the effect and use of medicines are studied in it.
Entomology	The science of insects.
Biometry	Biological measurement and statistics are used in it.
Biogeography	The geographical distribution of organisms on the Earth.
Surgery	Diseases are treated by means of operations.
Animal husbandry	Rearing animals, caring for them and gaining benefit from them.
Palaeontology	The science of fossils; the age of past life is known through the study of fossils.

3. Classification

Dividing different organisms into different groups or kingdoms on the basis of their characteristics is called **classification**.

4. Treatment of Diseases

Diseases are treated through both the science of medicines (pharmacology) and surgery. That is why the science of the treatment of diseases includes both of them.

5. Agriculture and Horticulture

Agriculture	Horticulture
In agriculture, crops are grown.	Horticulture is the part of agriculture in which new plant varieties are produced and the growth of fruits is cared for; new plant varieties, fruits and flowers are grown in it.

6. Fisheries

The work of breeding good fish and increasing fish production is called **fisheries**. Fish is a good source of protein, because fish contains a lot of protein.

7. Finding the Age of Fossils

Radio-dating and carbon dating are used to find out the **age** of fossils. Through palaeontology, the age of all organic life can be known.

8. Biology and Other Sciences

Biology is called a multidisciplinary subject because it is related to many other sciences, such as physics, chemistry, mathematics and geography, and it has many branches.

Biology linked with	Branch formed
Physics	Biophysics
Chemistry	Biochemistry
Mathematics	Biometry
Geography	Biogeography

Biology is also related to medicine, agriculture and other fields.

9. How Biology Serves Farmers

- By increasing crop production.
- By producing new plant varieties.
- By controlling pests.
- By improving animal rearing.

New plant varieties are produced through selection, grafting and genetic methods.

10. Contribution of Muslim Scientists

Muslim scientists made a great contribution to biology. Scientists of Europe also benefited from their books.

- **Jabir bin Hayyan** worked in chemistry.
- **Ibn Sina** worked in medicine.
- **Al-Razi** worked in skin diseases and smallpox-measles.
- **Al-Biruni** and others worked in botany and zoology.

1.2 Levels of Organisation

LESSON 2

1. Bio-elements: The Basis of Organisation

Bio-elements are the elements necessary for life, such as carbon, hydrogen, oxygen, nitrogen, phosphorus and sulphur. The elements necessary for life are found in all living things.

- The six basic elements are: carbon, hydrogen, oxygen, nitrogen, phosphorus and **sulphur**.
- A statement that leaves out sulphur from these six elements is incomplete, so it is a wrong statement.

The basis of life rests on **protoplasm**. Protoplasm is the chemical basis of life.

2. The Levels of Organisation

The levels of organisation run from small to large, and each larger level is made of the smaller levels:

Atom → molecule → cell → tissue → organ → system → organism → population → community → ecosystem → biosphere

- Similar cells together make a **tissue**.
- The correct order of biological organisation begins: atom → molecule → cell → tissue → organ.

3. Species, Population and Community

Level	Easy meaning
Species	A species is recognised when its members can interbreed and produce fertile offspring.
Population	A group of organisms of the same species living together in one place is called a population.
Community	Members of different species living together at a particular place form a community.
Biosphere	The part of the Earth where life is found is called the biosphere.

4. Population and Community: The Difference

Population	Community
A population is a group of members of one species.	In a community, members of different species live together.

5. Colonial and Multicellular Organisation

Colonial organisation	Multicellular organisation
Many cells live together but each cell does its own work, as in Volvox.	Cells together form tissues, organs and systems, and the work is divided.

- **Volvox** is a colony of green algae, so it is a multicellular body of green algae living in colonial organisation.
- The scientific name of the frog is **Rana tigrina**.

Solving a Biological Problem

UNIT 2

2.1 Biological Method

LESSON 3

1. Biological Method — Introduction

Biologists face many questions about living things. The problems on which biologists do not agree, and on which scientists debate, are called **biological problems**. To solve such problems in a proper, scientific way, biologists use the **biological method** — the way of solving biological problems by the scientific method.

The biological method follows a fixed sequence of steps: **problem → observation → hypothesis → reasoning → experiment → data collection → conclusion → theory → law**. So, in short, the order runs from hypothesis to reasoning, then to theory, and finally to law.

Step	Easy Meaning
Problem	The biological problem to be solved — a problem on which scientists debate
Observation	Looking at something carefully and collecting information about it; the solution of a biological problem begins with observation
Hypothesis	A testable statement which is checked and confirmed through experiment
Reasoning	Thinking about the hypothesis — inductive reasoning and deductive reasoning
Experiment	The practical test of the hypothesis
Data Collection	Collecting the results of the experiment; once data is in hand, its mathematical analysis is needed
Conclusion	The conclusion drawn from the experiment and the analysis of data
Theory	A tested explanation formed from tested conclusions
Law	A statement of a uniform and constant natural fact, tested again and again

2. Biological Problems, Observation and Hypothesis

The solution of every biological problem begins with **observation**. Observation means looking at something carefully and collecting information about it.

- **Quantitative observation:** In quantitative observation, **instruments** are used for numbers and measurement.

After observation comes the **hypothesis**. A hypothesis is a testable statement which is checked and confirmed — a statement that can be tested through experiment. The hypothesis is an important part of the scientific method.

An example from this unit: the cause of malaria is **Plasmodium**. A statement like this is first put forward as a hypothesis and is accepted only after it has been tested.

3. Reasoning — Inductive and Deductive

Reasoning is the step in which the hypothesis is thought through logically. There are two kinds of scientific reasoning:

- **Inductive reasoning:** Reasoning from the specific to the general. In inductive reasoning, a general conclusion is drawn from specific observations.
- **Deductive reasoning:** Reasoning based on "if-then" statements. In deductive reasoning, a specific conclusion is drawn from a general principle.

Inductive Reasoning	Deductive Reasoning
A general conclusion is drawn from specific observations	A specific conclusion is drawn from a general principle
Moves from the specific to the general	Usually works with "if-then" statements

4. Experiment, Data Collection and Conclusion

An **experiment** is needed so that the hypothesis can be tested. Without experiment, a hypothesis cannot become a theory — this is why the experiment is such an essential step of the biological method.

During and after the experiment, data is collected. Once data is in hand, **mathematical analysis** of the data is needed. The conclusion drawn from the experiment and the analysis of data is called the **derived conclusion**.

5. Theory and Law

- **Theory:** A theory is a tested explanation of a natural process. It is formed from tested conclusions. A theory is the most reliable scientific explanation because it has been tested again and again by experiments and it explains many observations.
- **Law:** A law is a statement of a uniform and constant natural fact, which has been tested again and again.

The unchanging answer to a uniform or constant natural fact is a law.

Theory	Law
A tested explanation of a natural process	A statement of a uniform and constant natural fact
A theory tells "why"	A law tells "what happens"

Remember: theory, law and hypothesis are all parts of the biological method. **Ratio** is not — ratio is a concept of mathematics. In the same way, **proportion** is not a part of the biological method; the method includes problem, observation, hypothesis, experiment and data collection.

6. Organising Data — Tables, Graphs, Ratio and Proportion

Data must be arranged properly before it can be understood. Both **tables and graphs** are useful for arranging data — with them, data can be seen in order, compared, and patterns can be easily understood.

- **Graph:** A graph is a way of showing data in picture form. The relationship between two variables is shown by means of a graph.
- **Ratio:** Ratio is the comparison of two quantities, such as two divided by four.
- **Proportion:** Proportion is an equation in which two ratios are equal. When two ratios are equal, that equation is a proportion.

7. Mathematics in Biology — Biomathematics and Mathematical Models

Mathematics plays an important role in solving biological problems. The science that studies biological systems by mathematical methods is called **biomathematics**.

A **mathematical model** is representing a biological system by mathematical equations and numbers. Biological science needs mathematical models so that data can be analysed, patterns can be understood and predictions can be made.

Remember: the biological method always moves in order — problem, observation, hypothesis, reasoning, experiment, data collection, conclusion, theory and law — and every hypothesis must pass the test of experiment before it can become a theory.

Biodiversity

UNIT 3

3.1 Introduction and Definition of Biodiversity

LESSON 4

1. Introduction to Biodiversity

Biodiversity is about the differences found in living organisms. It does not mean the sameness among organisms, but the differences that exist between them.

2. Definition of Biodiversity

The differences present among species are called **biodiversity**.

- A species is a distinct kind of organism, and different species differ from one another.
- When we talk about these differences among species as a whole, we use the term biodiversity.
- Biodiversity is therefore the name for variety and difference at the level of species.

Remember the definition in simple words: differences among species = biodiversity. This one definition is the base of this whole unit, because classification, the five kingdoms and conservation all start from the fact that species are different from one another.

3.2 Principles and Aims of Classification

LESSON 5

1. What Is Classification?

Classification is dividing organisms into groups on the basis of their characteristics.

The science of classification is called **taxonomy**.

2. Homologous and Analogous Structures

Type	Meaning	Role in Classification
Homologous structures	Structures that have the same internal structure, even though their functions (physiology) are different	Classification is based on homologous structures
Analogous structures	Structures that do the same work but have different internal structure	Not a basis for classification; in classification, all except analogous structures are involved

3. Taxonomic Hierarchy

Taxonomic hierarchy is the order of the ranks of classification. The correct order, from large to small, is:

Kingdom → Phylum → Class → Order → Family → Genus → Species

Rank	Meaning from the Source
Kingdom	The largest rank of classification, such as Monera, Protista, Fungi, Plantae and Animalia
Class	A large rank; in a list with order, genus and species, class brings all the others into one circle because it is the largest among them
Family	A rank smaller than order and larger than genus
Genus	A rank smaller than family and larger than species; it includes closely related species
Species	The group of organisms whose members can interbreed and produce fertile offspring

4. Purpose of Classification

- To identify organisms.
- To bring uniformity in their names.
- To understand the relationships among organisms.

Classification is therefore not just making groups. Its hierarchy, from kingdom down to species, helps us identify each organism and understand how organisms are related to one another.

3.3 History of Classification

LESSON 6

1. Classification into Three Kingdoms

One stage in the history of classification grouped organisms into three kingdoms. The chart of three kingdoms in the source is:

Kingdom	What It Contains
Protista	Unicellular eukaryotic organisms
Plantae	Plants
Animalia	Animals

2. The Four-Kingdom Stage and Kingdom Metaphyta

Classification later used four kingdoms. In the four-kingdom classification, Kingdom **Metaphyta** is the plant kingdom.

- Metaphyta includes bryophyta.
- Metaphyta includes gymnosperms.
- Metaphyta includes angiosperms.
- Metaphyta does **not** include algae.

So, in the history of classification, the number of kingdoms grew, and the groups placed inside each kingdom also changed. The key test point from this stage is simple: algae are the exception — they are not included in Kingdom Metaphyta, while bryophyta, gymnosperms and angiosperms are.

3.4 The Five Kingdoms

LESSON 7

1. How the Five Kingdoms Were Made

Organisms were classified into five kingdoms on the basis of their cell structure, method of obtaining food and body structure: Monera, Protista, Fungi, Plantae and Animalia.

Kingdom	Main Features	Example / Note
Monera	Prokaryotic, unicellular; no true nucleus	Bacteria; cyanobacteria are placed here because they are prokaryotic
Protista	Eukaryotic, mostly unicellular; has a true nucleus; many protists are unicellular	Amoeba
Fungi	Eukaryotic, heterotrophic, no chlorophyll, food taken by absorption; cell wall made of chitin	Mushroom
Plantae	Eukaryotic, autotrophic, has chlorophyll; cell wall made of cellulose	Plants
Animalia	Eukaryotic, heterotrophic, multicellular	Animals; multicellular animals are also called Metazoa

2. Important Differences

Comparison	First Group	Second Group
Monera vs Protista	Monera: prokaryotic, no true nucleus	Protista: eukaryotic, has a true nucleus
Fungi vs Plantae	Fungi: no chlorophyll, heterotrophic, chitin wall	Plantae: chlorophyll present, autotrophic, cellulose wall
Plantae vs Animalia	Plantae: make their own food (autotrophic), cells contain chlorophyll	Animalia: depend on others for food (heterotrophic), no chlorophyll

3. Fungi: Hyphae and Mycelium

- **Hyphae** are the thin threads of fungi.
- **Mycelium** is the network of hyphae (threads) of fungi.
- Fungi have no chlorophyll and take food by absorption, so organisms without chlorophyll that absorb food are fungi.

4. Why Amoeba Is Not in Animalia

Amoeba is not a member of the animal kingdom but of Protista. It has not been placed in the animal kingdom because it is unicellular and has no tissues and organs. So the pair Animalia → Amoeba is wrongly matched.

5. Where Are Viruses Placed?

Viruses are not placed in any of the five kingdoms. A virus is a non-cellular parasite, and it has not been placed in any kingdom because it is not cellular; it is neither living nor dead, but in between the two.

3.5 Binomial Nomenclature

LESSON 8

1. What Is Binomial Nomenclature?

Every species is given a double (binomial) scientific name. A scientific name is written by the binomial method:

- The first name is of the **genus**.
- The second name is of the **species**.

2. How a Scientific Name Is Written

Rule / Part	Detail
Number of names	Double — genus name + species name
Letters	Written in slanting (italic) letters
Dog	Canis lupus
Cat	Felis catus — the biological name of the cat

3. Why Is a Scientific Name Necessary?

A scientific name is necessary so that the same organism is known by the same name everywhere in the world and there is no confusion.

The correct way to write the scientific name of an organism is the binomial way, as in Canis lupus for the dog: genus first, species second, in slanting (italic) letters.

3.6 Conservation of Biodiversity

LESSON 9

1. What Affects Biodiversity?

Biodiversity is affected by all three of these:

- Deforestation
- Environmental damage
- Illegal hunting

2. Effect of Deforestation on Biodiversity

Deforestation destroys the homes of organisms, reduces the number of species, and many species fall into danger. That is why biodiversity decreases.

- Homes of organisms are destroyed.
- The number of species is reduced.
- Many species fall into danger.
- Result: biodiversity decreases.

3. Endangered Species

An **endangered species** is a species which faces the danger of permanent extinction. Animals which face the danger of permanent extinction are called endangered species.

Conservation of biodiversity therefore means protecting species from these dangers — deforestation, environmental damage and illegal hunting — so that species do not become endangered and biodiversity does not keep decreasing.

Cells and Tissues

UNIT 4

Microscopic Structure of the Cell

LESSON 10

1. What is a Microscope?

A microscope is the instrument with which small things can be seen magnified. In a microscope the visible image is magnified, so very small structures such as cells can be studied.

2. Magnification

Magnification is making the image of an object larger than the original in a microscope. In a light microscope the combination of the eyepiece lens and the objective lens gives a magnification of about 1000 times.

3. Resolution

Resolution is the smallest distance between two close points in a microscope that can still be seen separately. The smallest distance between two points is called resolution.

4. Light Microscope and Electron Microscope

A microscope has two main types. The light microscope uses light and its magnification is about 1000 times. The electron microscope uses an electron beam and its resolution and magnification are much greater.

- In the electron microscope the wavelength of the electron beam is short, so its resolution is greater.
- Ordinary light has a longer wavelength than electrons, so the electron microscope gives clearer images than an ordinary microscope.
- The short wavelength of the electron beam is responsible for the greater resolution of the electron microscope.

Feature	Light (Simple Compound) Microscope	Electron Microscope
Beam used	Light	Electron beam
Magnification	About 1000 times	Much greater
Resolution	Less	Much greater, because the electron beam has a short wavelength
Images	Less clear	Clearer images

In short, the electron microscope is different from the simple compound microscope because it uses an electron beam instead of light, and its resolution and magnification are much greater.

History of the Cell Theory

LESSON 11

1. Cell Theory

The postulates of cell theory are: all living things are made of cells, the cell is the basic unit of life, and new cells arise from pre-existing cells. The statement that the material of the cell membrane has no hereditary material is not a postulate of cell theory.

The nucleus inside the cell holds hereditary information and controls all the activities of the cell.

2. Prokaryotic and Eukaryotic Cells

Feature	Prokaryotic Cell	Eukaryotic Cell
True nucleus	No true nucleus	Has a true nucleus
Membrane-bound organelles	No membrane-bound organelles	Has membrane-bound organelles

3. Cell Wall

In plants, the strong fibres of the cell wall are made of cellulose. The secondary cell wall is made of cellulose and lignin.

4. Cell Membrane and the Fluid Mosaic Model

The cell membrane is made of a phospholipid bilayer and proteins. According to the fluid mosaic model, the cell membrane is made of a phospholipid bilayer in which pieces of protein float, like things floating in water. Proteins and carbohydrates can change place in the membrane, so the membrane is flexible and moving.

The cell membrane is semi-permeable because some substances can pass through it and some cannot, which keeps the inside of the cell controlled.

5. Nucleus

The nucleus is the control centre of the cell. Around it is a double membrane with pores. Inside are nucleoplasm, nucleolus and chromosomes. Nucleoplasm is the fluid inside the nucleus in which chromosomes and the nucleolus are present. The nucleus controls all the activities of the cell and holds hereditary information.

6. Endoplasmic Reticulum, Ribosomes and Golgi Body

- The rough endoplasmic reticulum makes proteins, and the smooth endoplasmic reticulum makes lipids and steroids.
- Ribosomes make proteins, not steroids; steroids are made in the smooth endoplasmic reticulum.
- Cisternae are the flattened sacs of the endoplasmic reticulum and the Golgi body.
- The correct sequence for a protein is: Ribosome, then endoplasmic reticulum, then Golgi body, then lysosome. A protein is made in the ribosome, then goes from the endoplasmic reticulum to the Golgi body and then to the lysosome.
- Vesicles are small sacs in the cell in which substances are enclosed and carried from one place to another.

7. Cell Organelles: Lysosomes, Mitochondria and Chloroplasts

Organelle	Main Work	Where Found / Note
Lysosome	Contains digestive enzymes which help in digestion inside the cell and break things down	Small organelles of animal cells that help in intracellular digestion
Ribosome	Makes proteins	Does not make steroids
Mitochondrion	Produces energy in the form of ATP through respiration; called the powerhouse of the cell	Present in all eukaryotic cells
Chloroplast	Makes food with light energy by photosynthesis	Present only in plant cells

The mitochondrion is called the powerhouse of the cell because in it energy is produced in the form of ATP through respiration.

8. Centrioles

The microscopic tubules arranged in a special pattern to form centrioles are 27 in number, that is 9 groups of three.

9. Movement Into and Out of the Cell

Osmosis, diffusion and active transport all three cause the movement of substances into and out of the cell.

- Diffusion is a passive process in which no energy is spent.
- Diffusion, facilitated diffusion and osmosis are passive processes in which no energy is spent; active transport uses energy, so active transport is the odd one out among them.
- Facilitated diffusion is a special type of diffusion of specific substances through carrier proteins. In facilitated diffusion substances move from higher to lower concentration through carrier proteins and no energy is spent.
- In active transport substances move from lower to higher concentration and energy is spent.
- Exocytosis is the process in which things from inside the cell are pushed out of the cell by means of vesicles.
- When a plant cell loses water, the cytoplasm shrinks, and this process is called plasmolysis.

Process	Direction	Energy	Special Point
Facilitated diffusion	Higher to lower concentration	No energy spent	Through carrier proteins
Active transport	Lower to higher concentration	Energy spent	Uses energy, unlike passive processes

Cell Size and Shape in Relation to Surface Area and Volume

LESSON 12

1. The Cell as the Structural and Functional Unit

The cell of organisms is called the structural and functional unit. It is called structural because all organisms are made of cells, and it is called functional because all the activities of life take place in the cell itself.

2. Activity: Looking at Plant Cells Under the Microscope

Onion peel is used to observe plant cells under the microscope. Iodine is used for onion peel because iodine stains the nucleus and other parts, so they can be seen clearly in the microscope.

- Iodine stains the nucleus and other parts of the cell.
- Because of this staining, the parts can be seen clearly in the microscope.

Note: This lesson is short in the English source pairs. Only the onion-peel activity and the idea of the cell as the structural and functional unit are given in the source for this topic, so no extra facts about cell sizes or surface area and volume have been added here.

Animal and Plant Tissues

LESSON 13

1. What is a Tissue?

A tissue is a group of cells of the same kind which together do a particular job.

- A tissue is a group, not a single cell.
- The cells in a tissue are of the same kind.
- Together, these cells do a particular job.

In other words, cells of the same kind join together and work together for one particular job, and this group is called a tissue. Animal and plant bodies are built from such groups of cells working together.

Note: This lesson is short in the English source pairs. Only the definition of a tissue is given in the source for this topic, so no extra lists of tissue types have been added here.

Cell Cycle

UNIT 5

Chromosomes

LESSON 14

1. What Are Chromosomes?

Chromosomes are thread-like structures. They appear in the cell at the time of cell division.

2. Parts of a Chromosome

A chromosome consists of two halves called chromatids. The two chromatids are joined to each other at a point called the centromere.

- **Chromatids:** the two halves of a chromosome, joined at the centromere.
- **Centromere:** the point (part) of the chromosome where the two chromatids are joined to each other.

3. Sister Chromatids and Homologous Chromosomes

- Two chromatids that are genetically identical and copies of each other are called **sister chromatids**.
- Chromosomes in a cell that are similar in arrangement to each other are called **homologous chromosomes**.

4. Key Terms

Term	Meaning
Chromosome	Thread-like structure that appears at the time of cell division
Chromatids	The two halves of a chromosome joined at the centromere
Centromere	The point where two chromatids are joined
Sister chromatids	Two chromatids that are genetically identical copies of each other
Homologous chromosomes	Chromosomes similar in arrangement to each other
Acrocentric chromosome	A chromosome in which one arm is very short and the other arm is very long

The Cell Cycle

LESSON 15

1. What Is the Cell Cycle?

The series of changes between one cell division and the next is called the **cell cycle**. It is the full sequence a cell passes through from one division to the next division.

2. Interphase — the Growth Stage

Interphase is the stage in which the cell grows and enzymes are made. It is called the growth process because during it:

- The cell grows in size.
- Enzymes are made.
- DNA is copied.
- Preparation for division takes place.

3. Cytokinesis — Division of the Cytoplasm

Cytokinesis is the division of the cytoplasm of the cell. By cytokinesis the cell finally divides into two parts, completing one turn of the cell cycle.

Term	Meaning
Cell cycle	The series of changes between one cell division and the next
Interphase	The stage in which the cell grows and enzymes are made; DNA is copied and the cell prepares for division
Cytokinesis	The division of the cytoplasm of the cell

Mitosis

LESSON 16

1. What Is Mitosis For?

Mitosis is involved in **growth, repair and asexual reproduction**. Mitosis is needed for growth because new cells are formed by it and the body grows.

2. Stages of Mitosis

Stage	What happens
Prophase	Chromosomes appear and the nuclear membrane disappears. In animal cells the centrioles separate and move towards the poles.
Metaphase	The chromosomes come in a row and line up at the equator of the cell.
Anaphase	The chromatids separate and move towards the poles.
Telophase	The chromatids reach the poles and their movement stops. The nuclear membrane forms again (reappears).

After these stages, by **cytokinesis** the cytoplasm divides and the cell divides into two parts.

3. Points to Remember

- In telophase the nuclear membrane **reappears** — saying it disappears in telophase is wrong; it disappears in prophase.
- The stage in which chromatids reach the poles and their movement stops is telophase.
- Separation of centrioles in animal cells takes place in prophase.

Apoptosis and Necrosis — the Two Paths of Cell Death

LESSON 17

1. Two Paths of Cell Death

Cells can die in two different ways: **apoptosis** and **necrosis**.

2. Apoptosis — Organised Self-Destruction

Apoptosis is the sequence of changes by which a cell kills itself. It is the process in which a cell ends itself in an organised way — the organised self-destruction of a cell.

3. Necrosis — Death by External Factors

Necrosis is the death of cells caused by external factors. Unlike apoptosis, it is a disorderly death, not a planned one.

4. Apoptosis vs Necrosis

Point	Apoptosis	Necrosis
What it is	The organised self-destruction of a cell; the cell kills itself	The death of cells caused by external factors
Manner	Organised and planned	Disorderly
Cause	The cell's own sequence of changes	External factors acting on the cell

Meiosis — Reduction Division

LESSON 18

1. What Is Meiosis?

Meiosis is the type of cell division in which the chromosome number becomes half. In organisms, meiosis produces **gametes**. A spore mother cell also produces spores by meiosis. Meiosis does **not** produce somatic (body) cells.

2. Why Meiosis I Is Called Reduction Division

Meiosis I is called reduction division because in it the number of chromosomes becomes half. The chromosome number stays constant generation after generation because in meiosis the number becomes half and in fertilisation it becomes full again.

3. Two Divisions of Meiosis

Division	Stages
Meiosis I	Prophase I, metaphase I, anaphase I, telophase I
Meiosis II	Prophase II, metaphase II, anaphase II, telophase II

In the end, four gametes are formed with half the number of chromosomes. No interphase occurs between meiosis I and meiosis II, because there is no need to copy DNA again; the chromosomes go directly into the second division.

4. Sub-stages of Prophase I

Prophase I has five sub-stages: leptotene, zygotene, pachytene, diplotene and diakinesis.

Term / Sub-stage	What happens
Bivalent	A pair of homologous chromosomes formed in meiosis
Pachytene	Chromosomes thicken and crossing over takes place

Term / Sub-stage	What happens
Chiasmata	The points where homologous chromosomes meet each other and crossing over takes place
Diakinesis	The last stage of prophase I; chromosomes thicken further and the nuclear membrane disappears

5. More Events of Meiosis

- In metaphase I, homologous chromosomes arrange themselves in pairs at the equator.
- In anaphase II, centromeres divide and sister chromatids separate and move towards the poles.
- Prophase I differs from prophase II: in prophase I homologous chromosomes form pairs and crossing over takes place; in prophase II no crossing over takes place and chromosomes are single.
- When homologous chromosomes fail to separate, it is called **non-disjunction**.

Enzymes

UNIT 6

Definition of Enzymes

LESSON 19

1. Metabolism

The chemical reactions taking place in living organisms are together called metabolic reactions. Metabolism includes both kinds of reactions.

Type	Meaning
Anabolism (constructive reactions)	Large molecules are formed from small molecules, and energy is spent.
Catabolism (destructive reactions)	Large molecules break down into small molecules, and energy is released.

Metabolic processes are of two types: anabolism and catabolism.

2. Conditions Needed for Chemical Reactions

Chemical reactions need specific conditions so that they can proceed correctly. Those conditions are:

- Temperature
- pH

3. Activation Energy

Activation energy is the minimum energy needed to start a reaction.

Enzymes lower the activation energy of a chemical reaction. Enzymes bind with the substrate and lower the energy needed to start the reaction, so the reaction happens quickly.

4. Definition of Enzymes

Enzymes are biological catalysts that speed up chemical reactions in the cell.

Most enzymes are made of proteins, so enzymes are proteins in nature.

Characteristics of Enzymes

LESSON 20

1. Properties of Enzymes

Enzymes are biological catalysts that speed up chemical reactions in the cell. Their properties are:

- Mostly made of proteins.
- Act on a specific substrate.
- Are not changed at the end of the reaction.
- A small amount is enough.
- Are sensitive to temperature and pH.
- Have no effect on the equilibrium of a reaction; they only speed up its rate.

The presence of enzymes also does not affect the amount of the end product; it only speeds up the rate of the reaction.

2. Important Terms

Term	Meaning
Substrate	The substance on which an enzyme acts.
Active site	The small part of the enzyme where the substrate binds.
Cofactor	The non-protein part of an enzyme which is necessary for the action of the enzyme.
Prosthetic group	A cofactor which is firmly attached to the enzyme.
Holoenzyme	A complete enzyme formed by the combination of apoenzyme and cofactor.

3. Activators and Inhibitors

Activator	Inhibitor
Activators are substances that increase the action of enzymes.	Inhibitors are substances that stop or reduce the action of enzymes.

Acting with inhibitors is not a property of enzymes; inhibitors reduce their action.

4. Factors Affecting the Action of Enzymes

Factor	Effect
Temperature	Action is best at a suitable temperature. As the temperature rises, the initial reaction rate increases, until the enzyme is damaged at very high temperature.
pH	Every enzyme has a suitable pH. The pH at which an enzyme works best is called its optimum pH. Extreme change in pH can cause denaturation of enzymes.
Amount of substrate	As the amount of substrate increases, the action of the enzyme speeds up, until all the enzymes are busy; after that the rate does not increase further.
Activators and inhibitors	Activators increase the action, and inhibitors reduce it.

pH, amount of substrate and temperature affect the action of enzymes; organic gradients do not.

5. Use of Enzymes in Factories

In factories, enzymes are used in making:

- Paper
- Bread
- Beverages
- Washing products

Mechanism of Enzyme Action

LESSON 21

1. How an Enzyme Works

A substrate is the substance on which an enzyme acts. The active site is the part of the enzyme where the substrate binds.

- The substrate binds at the active site of the enzyme.
- Enzymes bind with the substrate and lower the energy needed to start the reaction.
- Because the activation energy is lowered, the reaction happens quickly.
- The presence of enzymes does not affect the amount of the end product; it only speeds up the rate of the reaction.
- The enzyme itself is not changed at the end of the reaction.

2. Two Models of Enzyme Action

Model	Main idea
Lock and key model	Enzyme and substrate have specially matching geometric shapes, so they fit each other like a lock and a key.
Induced fit model	The active site continuously changes its shape until the substrate binds to it; the active site changes its shape according to the substrate.

In short: in the lock and key model the shapes already match, while in the induced fit model the active site changes its shape to fit the substrate.

Specificity of Enzymes

LESSON 22

1. Why Enzymes Are Specific

Enzymes are specific in nature because the active site of each enzyme fits only one particular substrate.

One property of enzymes is that they act on a specific substrate. Because the active site fits only that substrate, one enzyme does not act on other substances in the same way.

2. Examples of Specific Action

Enzyme	Substrate it acts on
Lipase	Lipids
Trypsin	Protein

Protein does not act on carbohydrate. That is why, in the pairing Protein to carbohydrate, this pair is the odd one out: lipase acts on lipids and trypsin acts on protein, but protein does not act on carbohydrate.

3. Many Enzymes in the Human Body

In the human body there are more than 1000 enzymes. Each one acts on its own specific substrate, which is possible because the active site of each enzyme fits only one particular substrate.

Bioenergetics

UNIT 7

7.1 Bioenergetics and the Role of ATP

LESSON 23

1. What is Bioenergetics?

Bioenergetics is the science in which the conversion and use of energy in organisms is studied. The conversion of free energy into different forms in organisms is also called bioenergetics.

2. Energy and Its Source

Energy is the ability to do work. The source of energy is only the sun — the sun is the original source of energy.

Plants use light energy to make use of simple inorganic molecules, water and carbon dioxide. In this way, energy from the sun enters living organisms through plants.

3. Food Chain

A food chain is the line of organisms in which one organism eats another and energy moves forward. Energy therefore passes from one organism to the next along the food chain.

4. Oxidation

Oxidation is the reaction in which electrons and protons are lost. The loss of electrons and protons is called an oxidation reaction.

5. ATP — The Treasure of Cellular Energy

The treasure of cellular energy is ATP. Energy in organisms is stored in a special molecule called ATP, and the energy obtained by breaking down a food molecule is stored in the form of ATP.

Point	Detail
Energy stored in	ATP — a special molecule in organisms
Energy in each mole of ATP	7.3 Kcal/mole
Original source of energy	The sun
Energy from breaking down food	Stored in the form of ATP

6. How Chemical Energy is Transferred

Chemical energy is stored in food molecules. In respiration, that energy is transferred into ATP, and then it is used in the activities of the cell.

- Chemical energy is stored in food molecules.
- In respiration, this energy is transferred into ATP.
- The ATP is then used for the activities of the cell.

7.2 Photosynthesis

LESSON 24

1. Photosynthesis — The Basic Molecule Formed

In photosynthesis, glucose is basically formed. Plants use light energy to make use of simple inorganic molecules, water and carbon dioxide, and glucose is the basic molecule formed during this process.

- Basic molecule formed: glucose
- Photosynthesis is the only process that produces free oxygen.
- In plants, glucose is converted into starch, cellulose, sucrose and other substances.

2. The Chloroplast

The chloroplast is a double-membrane organelle. Different pigments in the chloroplast absorb different light.

Part	Detail
Stroma	The semi-fluid protein liquid of the chloroplast, in which grana float
Grana	Stacks of thylakoids in the chloroplast
Thylakoid	The place where the light-dependent (light) reaction takes place
Pigments	Different pigments in the chloroplast absorb different light

3. The Two Stages of Photosynthesis

Light Reaction	Dark Reaction
Takes place in the thylakoid	Takes place in the stroma
Needs light	Does not need direct light
Light energy is converted into ATP and NADPH2	Carbon dioxide is converted into glucose with the help of ATP and NADPH2 (Calvin cycle)
Water breaks — photolysis	Glucose is formed

4. Photolysis

Photolysis is the breaking of water in light, from which oxygen, protons and electrons come out. The breaking of water in the presence of light is called photolysis.

5. Why is the Second Stage Called the Dark Reaction?

The second stage of photosynthesis is called the dark reaction because it does not need direct light; it uses the ATP and NADPH2 obtained from the light reaction.

6. How Carbon Dioxide and Temperature Affect the Rate

- As carbon dioxide increases, the rate of photosynthesis increases, until other factors run short.
- As temperature increases, the rate increases.
- At very high temperature, enzymes are damaged and the rate decreases.

7.3 Respiration

LESSON 25

1. What is Respiration?

Aerobic respiration is the process in which glucose breaks down in the presence of oxygen and a lot of energy (36 ATP) is obtained. The energy obtained by breaking down a food molecule is stored in the form of ATP.

2. Breathing and Cellular Respiration

Breathing	Cellular Respiration
The physical process of air coming in and going out	The chemical process of the breaking down of glucose in the cell, by which energy is obtained

3. Aerobic and Anaerobic Respiration

Aerobic Respiration	Anaerobic Respiration
Oxygen is used	Oxygen is not used
36 ATP are obtained	Only 2 ATP are obtained
Glucose breaks down in the presence of oxygen	Glucose breaks down without oxygen — fermentation, with a little energy obtained

4. Stages of Aerobic Respiration

Stage	Place	What Happens
(1) Glycolysis	Cytosol	Glucose breaks into pyruvic acid; a net of 2 ATP are formed
(2) Krebs cycle	Mitochondria	Pyruvic acid breaks down and carbon dioxide comes out
(3) Electron transport chain	Cristae of mitochondria	ATP are formed on the cristae

Total from aerobic respiration: 36 ATP from one mole of glucose.

5. Pyruvic Acid

Pyruvic acid is the substance formed at the end of glycolysis which goes into the Krebs cycle.

6. Energy Calculations

- One mole of glucose gives 2827 kJ of energy — this is the maximum energy obtained from one mole of glucose.
- For 14135 kJ of energy in the aerobic way: 14135 ÷ 2827 = 5 moles of glucose are used.
- One mole of glucose gives 36 ATP, so for 180 ATP: 180 ÷ 36 = 5 moles of glucose are used.

7. Fermentation

Fermentation is the process in which glucose breaks down without oxygen and a little energy is obtained.

Harm of lactic acid fermentation: In lactic acid fermentation, lactic acid collects, which causes tiredness and pain in the muscles.

8. Respiration and Photosynthesis — Difference

Respiration	Photosynthesis
Glucose breaks down and energy is released	Glucose is made with light energy
Oxygen is used and carbon dioxide comes out	The reverse happens — carbon dioxide is used and oxygen comes out

Nutrition

UNIT 8

8.1 Nutrition in Plants

LESSON 26

1. Types of Nutrition

Nutrition is the way organisms obtain food. Organisms that obtain their food from decayed matter are called **saprophytic**.

Autotrophic nutrition	Heterotrophic nutrition
Organisms make their own food, such as plants.	Organisms depend on others for food, such as animals.

2. Mineral Nutrition in Plants

A fertiliser is a substance in which the **nutrients** necessary for the food of plants are present. Fertilisers are necessary for plants because from them plants get nitrogen, phosphorus and other nutrients with which plants grow.

- **Nitrogen:** needed by plants because it is required for making proteins and chlorophyll.

3. Micro-nutrients and Macro-nutrients

Micronutrients	Macronutrients
Nutrients needed by the body in small amounts, such as vitamins and minerals.	Nutrients needed in large amounts, such as carbohydrates, proteins and fats.

4. Types of Fertilisers

Inorganic fertilisers	Organic fertilisers
Made in chemical factories and act quickly, but harm the environment.	Made from natural materials, such as dung, and improve the soil.

5. Environmental Hazards of Chemical Fertilisers

Excessive use of fertilisers **damages** the ecosystem. Excessive use of chemical fertilisers damages the soil, pollutes water, increases algae in lakes (eutrophication) and increases greenhouse gas in the air. That is why chemical fertilisers are dangerous for the environment.

6. Important Food Components

Component	Key point
Carbohydrates	Carbohydrates that are digested quickly are called simple carbohydrates.
Fat	Each gram of fat gives the body about 9 calories, which is double that of carbohydrates and proteins, so fat is called a more powerful food. Fat breaks into fatty acids and glycerol, not starch.
Protein	Seafood is a very good source of protein because it is high in protein and often low in fat.
Vitamins	Organic substances needed in small amounts. Vitamin A is involved in the eye's vision pigment and is necessary for the vision pigment (rhodopsin).

7. Types of Vitamins

Fat-soluble vitamins	Water-soluble vitamins
A, D, E and K — dissolve in fat and are stored in the body.	B and C — dissolve in water and are not stored in the body.

An animal that covers a long distance in migration benefits most from storing energy in the form of fat, because fat stores the most energy.

8.2 Balanced Diet in Relation to Age, Sex and Activity

LESSON 27

1. Balanced Diet

Food that fulfils **all the needs of the body** is called a **balanced diet**. The needs of the body change with age; the period of rapid growth and maturity is called **puberty**.

2. Vitamins

Vitamins are organic substances needed by the body in small amounts and necessary for health. Deficiency of vitamin C causes **scurvy**.

3. Malnutrition

Malnutrition is food in which the nutrients needed by the body are not obtained in full amount.

4. Effects of Malnutrition

- Growth stops in children
- Kwashiorkor and marasmus occur
- Immunity decreases
- Anaemia occurs
- Heart diseases, constipation and obesity occur

Starvation is the condition in which the body does not get the food it needs and the body becomes weak.
Obesity is the condition in which too much fat collects in the body.

5. Protein-Deficiency Conditions

Marasmus	Kwashiorkor
Caused by deficiency of energy (calories) and the child becomes very weak.	Caused by deficiency of protein and the body swells.

Protein deficiency in nutrition creates imbalance in the body's system, and this disease is kwashiorkor.

6. Mineral-Deficiency Diseases

- **Iron:** deficiency causes anaemia, and iron deficiency is also related to low immunity. Anaemia is the condition in which the amount of red blood cells or haemoglobin in the blood decreases.
- **Iodine:** deficiency causes goitre, which is the swelling of the thyroid gland in the neck.
- **Calcium:** deficiency makes bones weak and teeth bad. Green leafy vegetables, broccoli and walnuts are sources of calcium; red meat is not a good source of calcium.

8.3 Human Digestive System

LESSON 28

1. Digestion

Digestion is the process in which food breaks into small molecules so that the body can absorb it.

Mechanical digestion	Chemical digestion
Food is broken into small pieces by chewing and peristalsis.	Enzymes break food down chemically.

2. Parts of the Digestive System in Order

- **Mouth:** food is chewed into small pieces and mixed with saliva.
- **Pharynx:** the place where the trachea and oesophagus both meet.
- **Oesophagus:** food moves onwards from the pharynx.
- **Stomach:** food turns into chyme; pepsin breaks protein and acid kills germs.
- **Small intestine:** pancreatic juice, bile and intestinal juice break food down completely, and it is absorbed through the villi.
- **Intestines:** the remaining part of the digestive tract after digestion in the stomach and small intestine.

The rhythmic and sequential movement of muscles that moves food onwards is called **peristalsis**.

3. Action in the Mouth

Chewing is necessary for swallowing because food becomes small pieces, mixes with saliva, and swallowing and digestion become easy.

4. Action of the Stomach

- Food turns into **chyme**, the semi-liquid substance of half-digested food in the stomach.
- **Pepsin** starts the digestion of protein in the stomach.
- The acid kills germs.
- The stomach protects itself from its acidic environment with a thick layer of mucus which covers the wall of the stomach.

5. Action of the Small Intestine

- **Bile** emulsifies fat in the intestine: it divides fat into small drops so that enzymes can act on it.
- The juices collected from the pancreas contain different **enzymes**, such as lipase, amylase and trypsin.
- Pancreatic juice, bile and intestinal juice break food down completely.
- The digested food is absorbed through the **villi**.

8.4 Disorders of the Gut

LESSON 29

1. Constipation

Constipation is the condition in which the belly is not cleared and waste comes out with difficulty. Constipation is also listed among the effects of malnutrition, together with heart diseases and obesity.

2. Ulcer

An ulcer is a wound in the inner wall of the stomach or intestines.

3. How the Stomach Normally Protects Its Wall

The stomach protects itself from its acidic environment with a thick layer of mucus which covers the wall of the stomach. The ulcer is the wound that appears in this inner wall of the stomach or intestines.

Note: this lesson is short because the source material for this unit gives only two gut disorders in detail — constipation and ulcer.

Transport

UNIT 9

9.1 Transport in Plants

LESSON 30

1. The Root: Entry of Water and Minerals

Water and mineral salts enter the plant through the root. The outer layer of the root is the epidermis, and some of its cells grow out to form root hairs. The part of the root between the endodermis and the epidermis is called the cortex.

2. Root Hairs

Root hairs are very important for the plant because they increase the absorbing surface. By increasing the surface area, root hairs allow the root to absorb more water and minerals from the soil.

- Root hairs grow out from the epidermis of the root
- They increase the surface area for absorption
- More surface area means more water and minerals are absorbed

3. Osmosis

Osmosis is the movement of water molecules from higher concentration to lower concentration. More exactly, osmosis is the movement of water molecules through a semi-permeable membrane from a region of higher concentration to a region of lower concentration. Water enters the root hairs and moves through the cortex by this process.

Term	Meaning
Cortex	The part of the root between the endodermis and the epidermis
Root hairs	Outgrowths of the epidermis that increase the absorbing surface area
Osmosis	Movement of water molecules through a semi-permeable membrane from higher to lower concentration

9.2 Transpiration Process

LESSON 31

1. What is Transpiration?

Transpiration is the loss of water from the inner parts of plants in the form of vapour. This loss of water as vapour is also simply called water loss. Only a small part of the water absorbed by the roots is used by the plant; the rest escapes as vapour, mostly through the leaves.

2. Stomata and Guard Cells

Stomata are small pores on the surface of leaves through which gases and water vapour pass. The opening and closing of stomata occurs with the help of guard cells, the special cells that surround each stoma.

- Stomata are small pores on the leaf surface
- Gases and water vapour pass through stomata
- Guard cells control the opening and closing of stomata

3. The Dorsiventral Leaf

A dorsiventral leaf is a leaf in which there is a difference between the upper and lower surfaces. Most broad leaves are dorsiventral, and their stomata are mostly on the lower surface.

4. Mechanism of Transpiration

The mechanism works as a sequence: the stomata open, water evaporates from the leaf cells into the air, and this loss creates a pull in the xylem. This transpiration pull draws more water upward from the roots.

5. Why Plants Need Transpiration

- By transpiration, water and minerals rise up through the plant
- The plant stays cool
- Food is transported through the plant

6. Factors Affecting Transpiration

Factor	Effect on Transpiration
Temperature	Higher temperature increases the rate of transpiration
Humidity	Higher humidity decreases the rate of transpiration
Wind	Wind carries the vapour away and increases the rate
Atmospheric pressure	Lower atmospheric pressure increases the rate of transpiration

9.3 Transport of Water and Food in the Stem

LESSON 32

1. The Two Transport Tissues

In flowering plants, transport through the stem is carried out by two tissues. Xylem carries water and minerals from the roots to the leaves. Phloem carries food (sucrose) from the leaves to other parts of the plant. Organic substances (food) are transported through the phloem.

2. Structure of Xylem

In flowering plants, xylem is made of four types of tissues:

- Tracheids
- Vessels
- Xylem parenchyma
- Xylem fibres

3. How Water Moves in the Xylem

Water rises in the xylem by three forces working together: the pull of transpiration of water from the leaf, the sticking of water molecules to each other, and root pressure. The transpiration pull is the main force that draws the water column upward.

4. Structure of Phloem: The Sieve Plate

A sieve plate is the wall with pores between the tubes of phloem through which food passes. Food solution moves from one sieve tube to the next through these pores.

5. Source and Sink

Phloem transport of sucrose can be described as a moving mechanism from "source" to "sink". The source is the part where food is made, such as the leaf. A sink is the part of the plant where food is stored or used. The sink can be both a storage organ and a growing part of the plant.

Xylem	Phloem
Carries water and minerals	Carries food (sucrose)
From roots to leaves	From leaves (source) to storage or growing parts (sink)
Made of tracheids, vessels, xylem parenchyma and xylem fibres	Contains sieve tubes joined by sieve plates

9.4 Transport in Animals

LESSON 33

1. Why Animals Need a Transport System

In multicellular animals, many cells lie far from the body surface, so simple diffusion cannot supply them. Animals therefore have a circulatory system in which blood carries food, oxygen and waste materials to and from the cells of the body.

2. Open Circulatory System

In an open circulatory system, blood flows in open spaces outside vessels. The blood leaves the vessels and bathes the tissues directly in open spaces before returning to the heart.

3. Closed Circulatory System

In a closed circulatory system, blood always stays in vessels. The type of circulatory system in which blood flows through vessels is called the closed circulatory system. Man and other mammals have a closed circulatory system.

Open Circulatory System	Closed Circulatory System
Blood flows in open spaces outside vessels	Blood always stays inside vessels
Blood comes into direct contact with the tissues in open spaces	Blood remains within arteries, veins and capillaries

Because the blood stays inside vessels in a closed system, it can be pumped around the body quickly and under pressure, which suits large and active animals.

9.5 Transport in Man

LESSON 34

1. Blood: The Transport Fluid

Blood is a red fluid which transports food, oxygen and waste in the body. It has two main parts: plasma (55%), which contains water, proteins and nutrients; and corpuscles, the blood cells.

2. Composition of Blood

Component	Function / Feature
Plasma (55%)	The fluid part; contains water, proteins and nutrients
Red blood cells (RBC)	Carry oxygen; have no nucleus
White blood cells (WBC)	Protect against diseases; have a nucleus
Platelets	Clot the blood

3. Plasma Proteins and Granulocytes

Human plasma proteins include fibrinogen and albumin. Haemoglobin is not a plasma protein; it is found inside the red blood cells. Granulocytes are a type of white blood cells which contain granules. The type of cancer affecting the bone marrow is called leukaemia.

4. Structure of the Heart

The heart is the pumping organ of the circulatory system and is protected in the pericardium. It is divided into four chambers, and blood flows through them in a fixed sequence: deoxygenated blood returning to the heart first flows into the right atrium, then passes to the right ventricle, which pumps it to the lungs; oxygenated blood returns to the left atrium, then passes to the left ventricle, which pumps it to the whole body. The heart has tricuspid, bicuspid and semilunar valves. Semilunar valves are valves that let blood flow in one direction only and stop it from flowing back.

- Right atrium — receives deoxygenated blood from the body
- Right ventricle — pumps deoxygenated blood to the lungs
- Left atrium — receives oxygenated blood from the lungs
- Left ventricle — pumps oxygenated blood to the body

5. The Two Circuits

In the pulmonary circuit, blood goes from the heart to the lungs and from the lungs back to the heart. In the systemic circuit, blood goes from the heart to all the organs of the body and back to the heart.

6. Heartbeat and Blood Pressure

The rhythmic pumping of blood by the heart throughout the body is called the heartbeat (pulse). Blood pressure is the pressure which blood exerts on the walls of the vessels.

7. Blood Vessels

Arteries	Veins	Capillaries
Carry blood from the heart to the organs; walls are thick	Carry blood from the organs to the heart; walls are thin	Made of a single layer of endothelium so substances can easily pass through them

8. Atherosclerosis

In atherosclerosis, fat collects in the vessels and the flow of blood stops. A blockage in the brain causes a stroke, and a blockage in the heart causes a heart attack.